

## 1. Logic

### Let us Recall

- The converse, inverse and contrapositive of the implication  $p \rightarrow q$  are:

Converse :  $q \rightarrow p$

Inverse :  $\sim p \rightarrow \sim q$

Contrapositive :  $\sim q \rightarrow \sim p$

- Quantifiers and quantified Statements : Look at the following statements:

$p$  : "There exists an even prime number in the set of natural numbers".

$q$  : "All natural numbers are positive".

Each of them asserts a condition for some or all objects in a collection. Words "there exists" and "for all" are called quantifiers. "There exists" is called existential quantifier and is denoted by symbol  $\exists$ . "For all" is called universal quantifier and is denoted by  $\forall$ . Statements involving quantifiers are called quantified statements. Every quantified statement corresponds to a collection and a condition. In statement  $p$  the collection is 'the set of natural numbers' and the condition is 'being even prime'.

What is the condition in the statement  $q$  ?

A statement quantified by universal quantifier  $\forall$  is true if all objects in the collection satisfy the condition. And it is false if at least one object in the collection does not satisfy the condition.

A statement quantified by existential quantifier  $\exists$  is true if at least one object in the collection satisfy the condition. And it is false if no object in the collection satisfy the condition.

|                  |                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------|
| Idempotent Law   | $p \wedge p \equiv p, \quad p \vee p \equiv p$                                                     |
| Commutative Law  | $p \vee q \equiv q \vee p, \quad p \wedge q \equiv q \wedge p$                                     |
| Associative Law  | $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r \equiv p \wedge q \wedge r$                    |
|                  | $p \vee (q \vee r) \equiv (p \vee q) \vee r \equiv p \vee q \vee r$                                |
| Distributive Law | $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$                                        |
|                  | $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$                                          |
| De Morgan's Law  | $\sim(p \wedge q) \equiv \sim p \vee \sim q, \quad \sim(p \vee q) \equiv \sim p \wedge \sim q$     |
| Identity Law     | $p \wedge T \equiv p, \quad p \wedge F \equiv F, \quad p \vee F \equiv p, \quad p \vee T \equiv T$ |
| Complement Law   | $p \wedge \sim p \equiv F, \quad p \vee \sim p \equiv T$                                           |

|                   |                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| Absorption Law    | $p \vee (p \wedge q) \equiv p, p \wedge (p \vee q) \equiv p$                                                          |
| Conditional Law   | $p \rightarrow q \equiv \sim p \vee q$                                                                                |
| Biconditional Law | $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p) \equiv (\sim p \vee q) \wedge (\sim q \vee p)$ |

**Ex. (1)** Write the converse, inverse, contrapositive and the negation of the implication: "If two sides of a triangle are congruent then it's two angles are congruent."

**Solution :**

Converse : If two angles of a triangle are congruent then it's two sides are congruent.

Inverse : If two sides of a triangle are not congruent then it's two angles are not congruent.

Contrapositive : If two angles of a triangle are not congruent then it's two sides are not congruent.

Negation : Two sides of a triangle are congruent but it's two angles are not congruent.

**Ex. (2)** Write (a) truth values and (b) negations of the following statements :

- i)  $\forall x \in R, x^2$  is positive. ii)  $\exists x \in R, x^2$  is not positive.  
 iii) Every square is a rectangle. iv) Some parallelograms are rectangles.

**Solution :**

a) Truth values

i) false because the square of 0 is not positive

ii) true because the square of 0 is not positive

iii) true iv) true

b) Negations

i)  $\exists x \in R, x^2$  is not positive. ii)  $\forall x \in R, x^2$  is positive.

iii) There exists a square which is not a rectangle.

iv) No parallelogram is a rectangle.

**Ex. (3)** Without using truth table prove that  $\{[(p \vee q) \wedge \sim p] \rightarrow \sim q\} \equiv q \rightarrow p$

**Solution :**

$$\text{L.H.S.} = \{[(p \vee q) \wedge \sim p] \rightarrow \sim q\}$$

$$\equiv \sim[(p \vee q) \wedge \sim p] \vee \sim q \quad (\dots \text{Conditional law} \dots)$$

$$\begin{aligned}
&\equiv [\sim(p \vee q) \vee p] \vee \sim q && (\text{De-Morgan's Law}) \\
&\equiv [(\sim p \wedge \sim q) \vee p] \vee \sim q && (\text{De-Morgan's Law}) \\
&\equiv [(\sim p \vee p) \wedge (\sim q \vee p)] \vee \sim q && (\text{Distributive Law}) \\
&\equiv [(t) \wedge (\sim q \vee p)] \vee \sim q && (\text{Complement Law}) \\
&\equiv (\sim q \vee p) \vee \sim q && (\text{Identity Law}) \\
&\equiv \sim q \vee p \vee \sim q && (\text{Associative Law}) \\
&\equiv \sim q \vee p && (\text{Idempotent Law}) \\
&\equiv q \rightarrow p && (\text{Conditional Law})
\end{aligned}$$

L. H. S. = R. H. S.

**Ex. (4)** Using truth table prove that  $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$ .

**Solution :**

| I | II | III      | IV       | V                     | VI                          | VII               | VIII              | IX                                         |
|---|----|----------|----------|-----------------------|-----------------------------|-------------------|-------------------|--------------------------------------------|
| p | q  | $\sim p$ | $\sim q$ | $p \leftrightarrow q$ | $\sim(p \leftrightarrow q)$ | $p \wedge \sim q$ | $q \wedge \sim p$ | $(p \wedge \sim q) \vee (q \wedge \sim p)$ |
| T | T  | F        | F        | T                     | F                           | F                 | F                 | F                                          |
| T | F  | F        | T        | F                     | T                           | T                 | F                 | T                                          |
| F | T  | T        | F        | F                     | T                           | F                 | T                 | T                                          |
| F | F  | T        | T        | T                     | F                           | F                 | F                 | F                                          |

From column (VI) and (IX) we conclude that

$$\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$$

**Ex. (5)** Is  $\sim(p \leftrightarrow q)$  equivalent to  $(\sim p) \leftrightarrow q$  ? Justify.

**Solution :**

| I | II | III      | IV                         | V                     | VI                          |
|---|----|----------|----------------------------|-----------------------|-----------------------------|
| p | q  | $\sim p$ | $\sim p \leftrightarrow q$ | $p \leftrightarrow q$ | $\sim(p \leftrightarrow q)$ |
| T | T  | F        | F                          | T                     | F                           |
| T | F  | F        | T                          | F                     | T                           |
| F | T  | T        | T                          | F                     | T                           |
| F | F  | T        | F                          | T                     | F                           |

From column no IV and VI we conclude that  $\sim(p \leftrightarrow q) \equiv (\sim p) \leftrightarrow q$

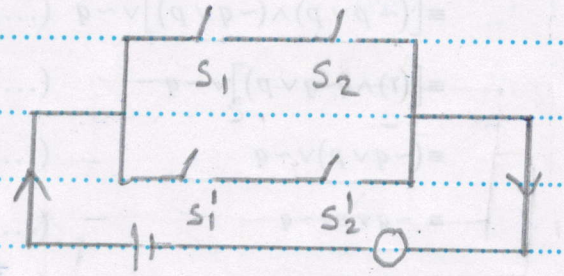
**Ex. (6)** Draw the switching circuits and prepare the input output tables for statement patterns :

a)  $(p \wedge q) \vee (\sim p \wedge \sim q)$

b)  $(p \vee q) \wedge \sim q$

**Solution :**

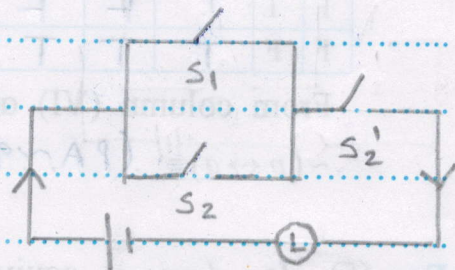
- ① Let p switch  $S_1$  is closed  
 q switch  $S_2$  is closed  
 $\sim p$   $S_1'$  is closed  
 $\sim q$   $S_2'$  is closed



| p | q | $p \wedge q$ | $\sim p$ | $\sim q$ | $(p \wedge q) \vee (\sim p \wedge \sim q)$ |
|---|---|--------------|----------|----------|--------------------------------------------|
| 1 | 1 | 1            | 0        | 0        | 1                                          |
| 1 | 0 | 0            | 0        | 1        | 0                                          |
| 0 | 1 | 0            | 1        | 0        | 0                                          |
| 0 | 0 | 0            | 1        | 1        | 1                                          |

②

| p | q | $\sim q$ | $p \vee q$ | $(p \vee q) \wedge \sim q$ |
|---|---|----------|------------|----------------------------|
| 1 | 1 | 0        | 1          | 0                          |
| 1 | 0 | 1        | 1          | 1                          |
| 0 | 1 | 0        | 1          | 0                          |
| 0 | 0 | 1        | 0          | 0                          |



**Ex. (7)** Using truth table prove that  $(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$

**Solution :**

| p | q | r | $p \wedge q$ | $(p \wedge q) \vee r$ | $p \vee r$ | $q \vee r$ | $(p \vee r) \wedge (q \vee r)$ |
|---|---|---|--------------|-----------------------|------------|------------|--------------------------------|
| T | T | T | T            | T                     | T          | T          | T                              |
| T | T | F | T            | T                     | T          | T          | T                              |
| T | F | T | F            | T                     | T          | T          | T                              |
| T | F | F | F            | F                     | T          | F          | F                              |
| F | T | T | F            | T                     | T          | T          | T                              |
| F | T | F | F            | F                     | F          | T          | F                              |
| F | F | T | F            | T                     | T          | T          | T                              |
| F | F | F | F            | F                     | F          | F          | F                              |

Sign of Teacher :

From column (v) and (viii) we conclude that

$$(p \wedge q) \vee r \equiv (p \vee r) \wedge (q \vee r)$$